

Genetic **Quality Control** & **Imputation** Procedure

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1. Genotype Quality Control

Before statistical calculations can begin, raw genotype array datasets must go through rigorous validation stages targeting both low-performing individual samples and sub-optimal variants.

Step 1 Sample-Level Quality Control

-  **Sample Call Rate Check:** Excludes individuals with elevated rates of missing genotypes (e.g., > 5% missingness). High missingness implies poor sample DNA quality or hybridization errors.
-  **Sex Check Verification:** Crosscheck genetic sex inferred from sex-chromosome homozygosity (F-coefficient) with recorded metadata to flag clinical mislabeling or sample mix-ups.
-  **Heterozygosity Outliers:** Detect deviations in expected sample heterozygosity. High rates indicate potential cross-sample contamination; abnormally low rates represent local inbreeding profiles.
-  **Duplicates & Relatedness:** Calculate identity-by-descent (IBD) or kinship coefficients to pinpoint duplicate submissions or close relatives, ensuring statistical independence in downstream analysis.

Step 2 Marker-Level (Variant) Quality Control

▼ SNP Call Rate Threshold

The proportion of samples for which a SNP **was successfully genotyped**. Variants lacking high validation are pruned to noise and bias downstream analyses (removing SNPs with $> 5\%$ missing data).

≡ Minor Allele Frequency (MAF)

Establishes the minimum frequency of a **less common allele at a given SNP**. Rare variants (typically $MAF < 1\%$ to 5%) are filtered out, as statistical power is insufficient to draw robust associative assumptions.

⚖ Hardy-Weinberg Equilibrium (HWE) Test

Checks whether the **observed genotype frequencies for a SNP match the expected frequencies in a stable, randomly mating population**. Severe deviations (e.g., $p < 1e-6$ in controls) signal technical error. Usually, HWE filtering is applied only to **controls**. For disease traits, this boundary is strictly limited to healthy controls to prevent filtering out actual causative mutations.

Step 3 Population Structure & Ancestry

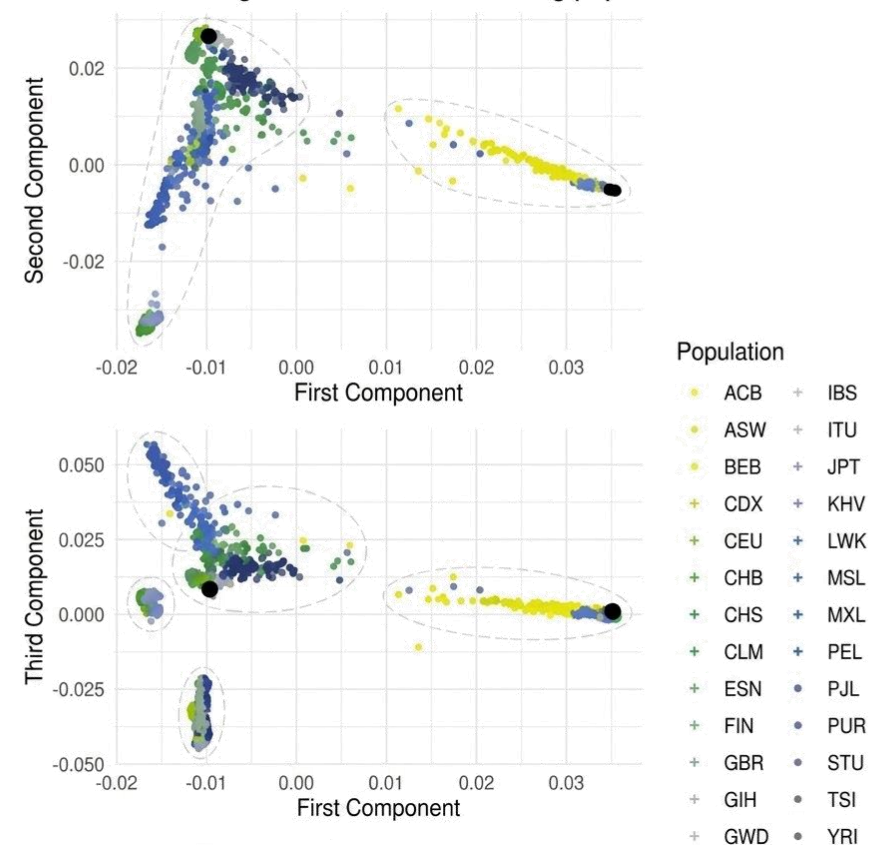
Population Stratification: Variations in allele frequencies caused by ancestral lineages rather than the disease phenotype of interest.

Principal Component Analysis (PCA): Helps identify clusters of individuals with similar ancestry to control for stratification bias.

Detecting Outliers: PCA can expose mislabeled ancestry records, cross-sample contamination, or batch effects driven by genotyping centers and chips.

Reference Projection: Can projects study cohort profiles onto established international datasets like the **1000 Genomes Project** to match populations accurately.

Principal component analysis (PCA) of 1000 genomes project, reference genome GRCh38, showing population structure



Resources Software Tooling



PLINK

The standard tool for filtering, call-rate calculations, HWE, and conversion.



KING & REAP

Specialized packages for evaluating biological kinship & complex relatedness.



EIGENSOFT

Calculates robust principal components for population stratification (PCA).



R Packages

Generates high-resolution visualization and custom statistical QC adjustments.

Hands-On Practice | QC Pipeline

Interactive Session: In this hands-on workshop track, we will run shell scripts to construct a standard genotyping QC workflow using terminal commands.

Using a sample dataset, we will execute sample-level missingness filters and calculate PCA matrices step-by-step.



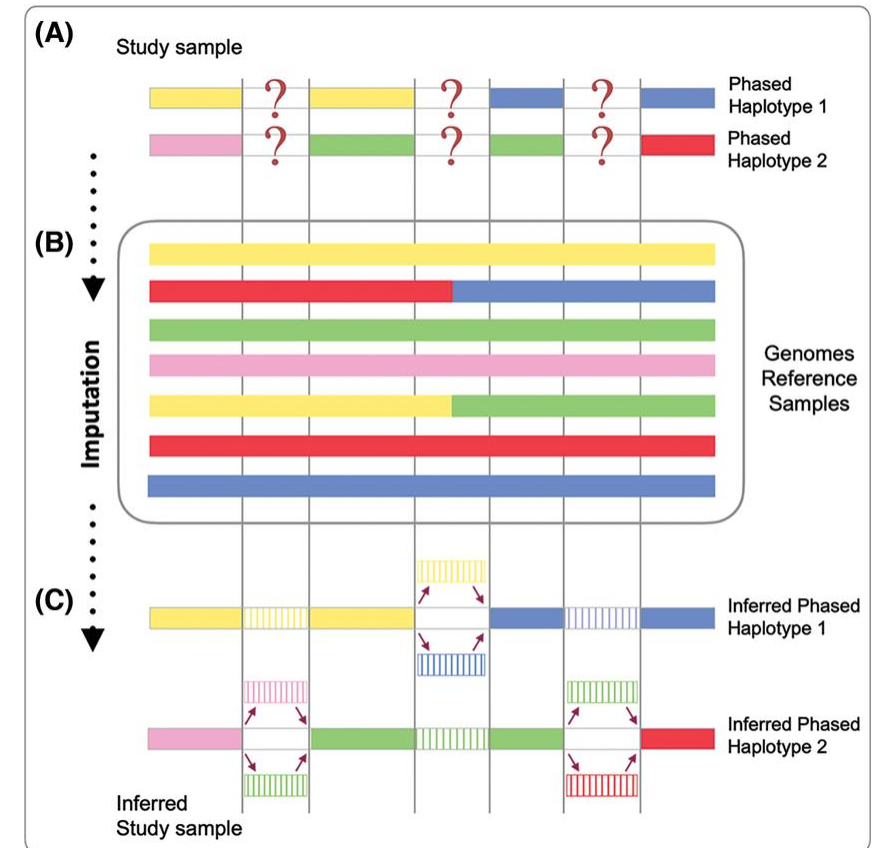
2. Genotype Imputation

Genotype Imputation uses statistical associations across known reference haplotypes to predict unobserved genotypes in microarray datasets.

Genotype Imputation Overview

- ➡ **Increases Genomic Coverage:** Fills in ungenotyped markers to approximate whole-genome data.
- ⚡ **Improves Study Power:** Enhances resolution for finding associations and identifying causal variants.
- 📦 **Reference Panels:** Uses **TOPMed** (97k diverse haplotypes) or **HRC** for high-density sequencing comparisons.

Critical: Successful imputation **requires rigorous pre-imputation QC** to ensure strand alignment and data quality.



Pre-Imputation Requirements Setup

✓ Standard Filters

Ensure that all basic sample and variant QC filters (HWE, MAF, and call rates) are completely applied before attempting alignment scripts.

🔗 Strand Alignment

Align raw dataset genotypes and forward strands with reference panels using Oxford's BIM-check tools to prevent incorrect allele mismatches.

(<https://www.chg.ox.ac.uk/~wrayner/tools/>)

✂ Autosome Isolation & Duplicates

- Prior to submission, convert chromosome layouts to distinct VCF formats per c.
- Make sure all duplicate SNPs at overlapping loci are systematically pruned out.
- Refer to the **Wrayner Strand Comparison Tools** for structural formatting templates.

Imputation Servers & Panels **Execution**

TOPMed Imputation Server: Uses the diverse, high-quality Trans-Omics for Precision Medicine reference panels, optimizing genetic resolution across non-European ancestral groupings.

Michigan Imputation Server 2: A free, high-speed, and secure platform utilizing Minimac4/phasing software with Haplotype Reference Consortium (HRC) maps.

Process Flow:

- Prepare and upload phased/sorted VCF files,
- Choose the appropriate matching reference panel,
- Set automated execution parameters,
- Download the high-density imputed files.



National Heart, Lung,
and Blood Institute

BioData CATALYST

TOPMed Imputation Server

<https://imputation.biodatacatalyst.nhlbi.nih.gov>

Michigan Imputation Server 2

Free Next-Generation Genotype Imputation Platform

<https://imputationserver.sph.umich.edu>

Post-Imputation Best Practices **Filtering**

 **Quality Filtering:** Filter based on accuracy metrics (e.g., R^2 or **INFO score** > 0.3 or 0.8).

 **Monomorphic Checks/Variant Exclusion:** Exclude poorly imputed variants and monomorphic SNPs appearing after imputation, which possess no statistical utility in GWAS.

 **Format Preparation:** Convert dosages back to PLINK or desired formats for association testing.

 **Documentation:** Record all parameters, versions, and reference panels used for reproducibility.

Hands-On Practice | Imputation Practice

Interactive Session: In this workshop session, we will configure, submit, and run datasets through the Michigan and TOPMed Imputation environments.

We will construct sorting commands to prep raw genomic assets for direct cloud integration.

